VPS Task -1

1. Create VPC with 2 private and 2 public subnets.

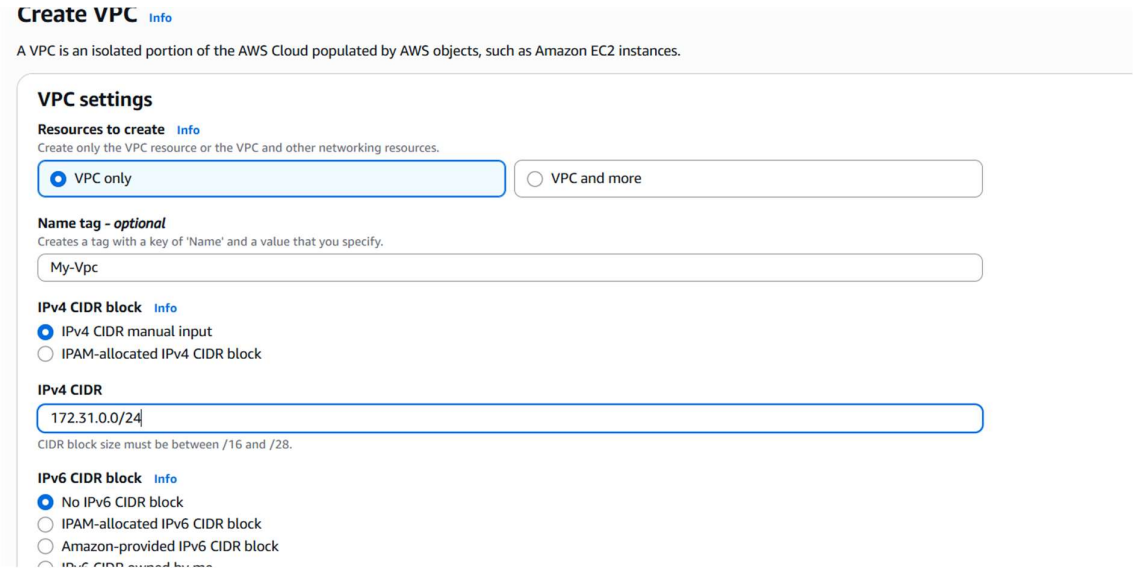
Step 1:- Open the VPC Console and create a new Vpc with new ip from B class and masking

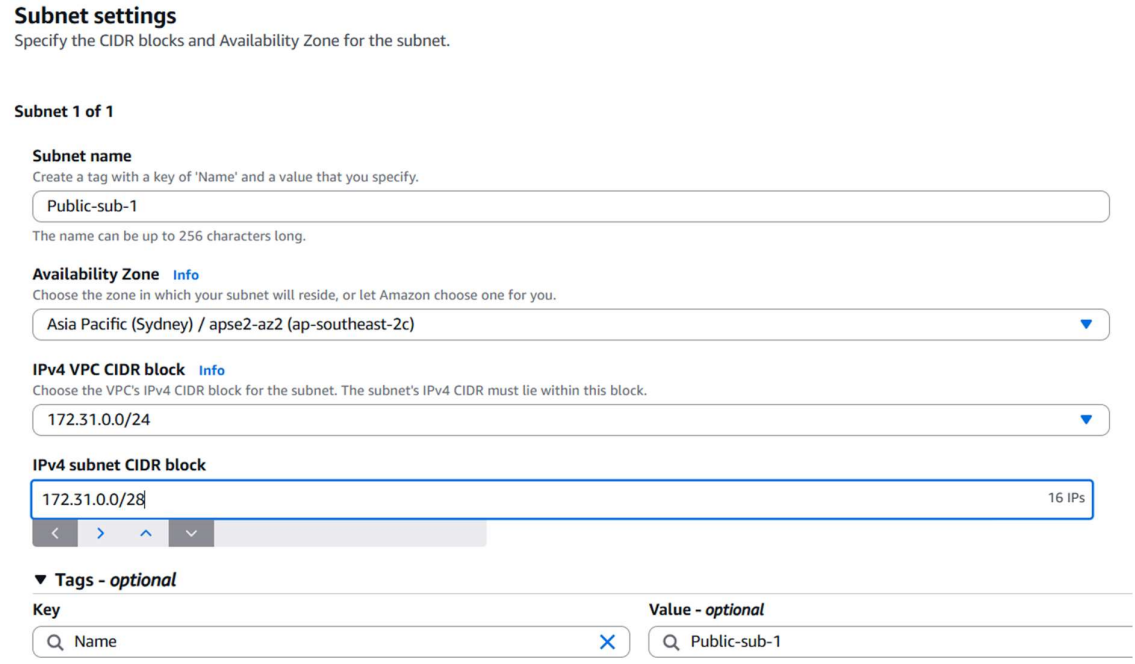
Step 2:- Now go to subnet and create new subnet

Step 3:- select new_Vpc and name the subnet then assign the IPv4 CIDR such that it does not exceed the VPC IPv4 CIDR

Step 4:- Similary create three more. Two for Public and private each

Step 5:- Private and public are just the names it will be assigned in internet gateway or Route table.





Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Public-sub-2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Sydney) / apse2-az2 (ap-southeast-2c)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.31.0.0/24

IPv4 subnet CIDR block
172.31.0.16/28 16 IPs

▼ **Tags - *optional***

Key	Value - *optional*

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-sub-1
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Sydney) / apse2-az2 (ap-southeast-2c)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.31.0.0/24

IPv4 subnet CIDR block
172.31.0.32/28 16 IPs

▼ **Tags - *optional***

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

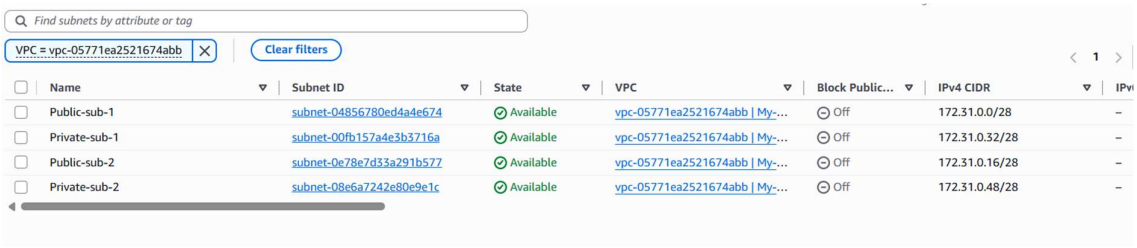
Subnet name
Create a tag with a key of 'Name' and a value that you specify.
Private-sub-2
The name can be up to 256 characters long.

Availability Zone Info
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
Asia Pacific (Sydney) / apse2-az2 (ap-southeast-2c)

IPv4 VPC CIDR block Info
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
172.31.0.0/24

IPv4 subnet CIDR block
172.31.0.50/28 16 IPs

▼ **Tags - *optional***

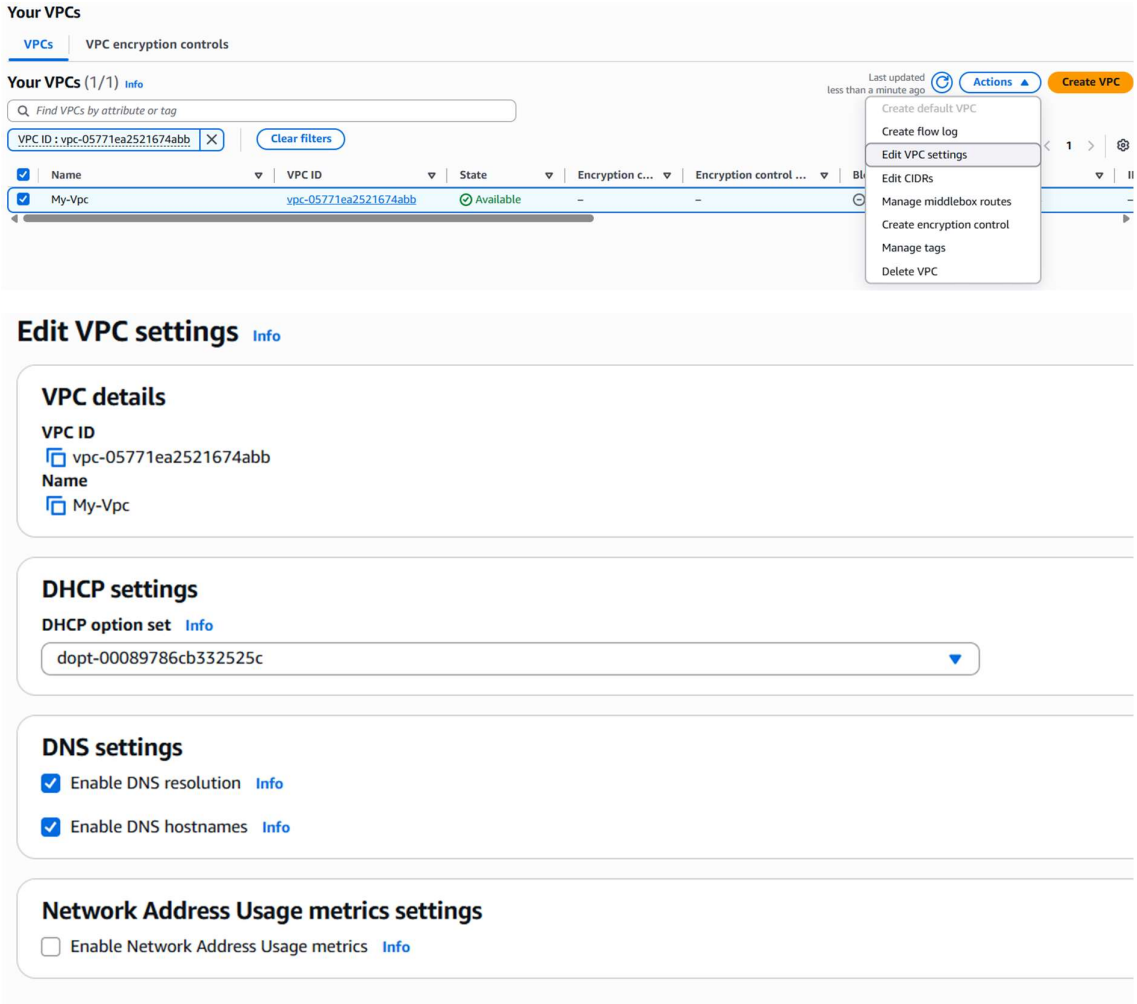


2. Enable DNS Hostname in VPC.

Step 1:- Select the VPC and Go to actions and choose VPC settings

Step 2:- Now we can see DNS Hostname enable it

Step 3:- Then save the changes.

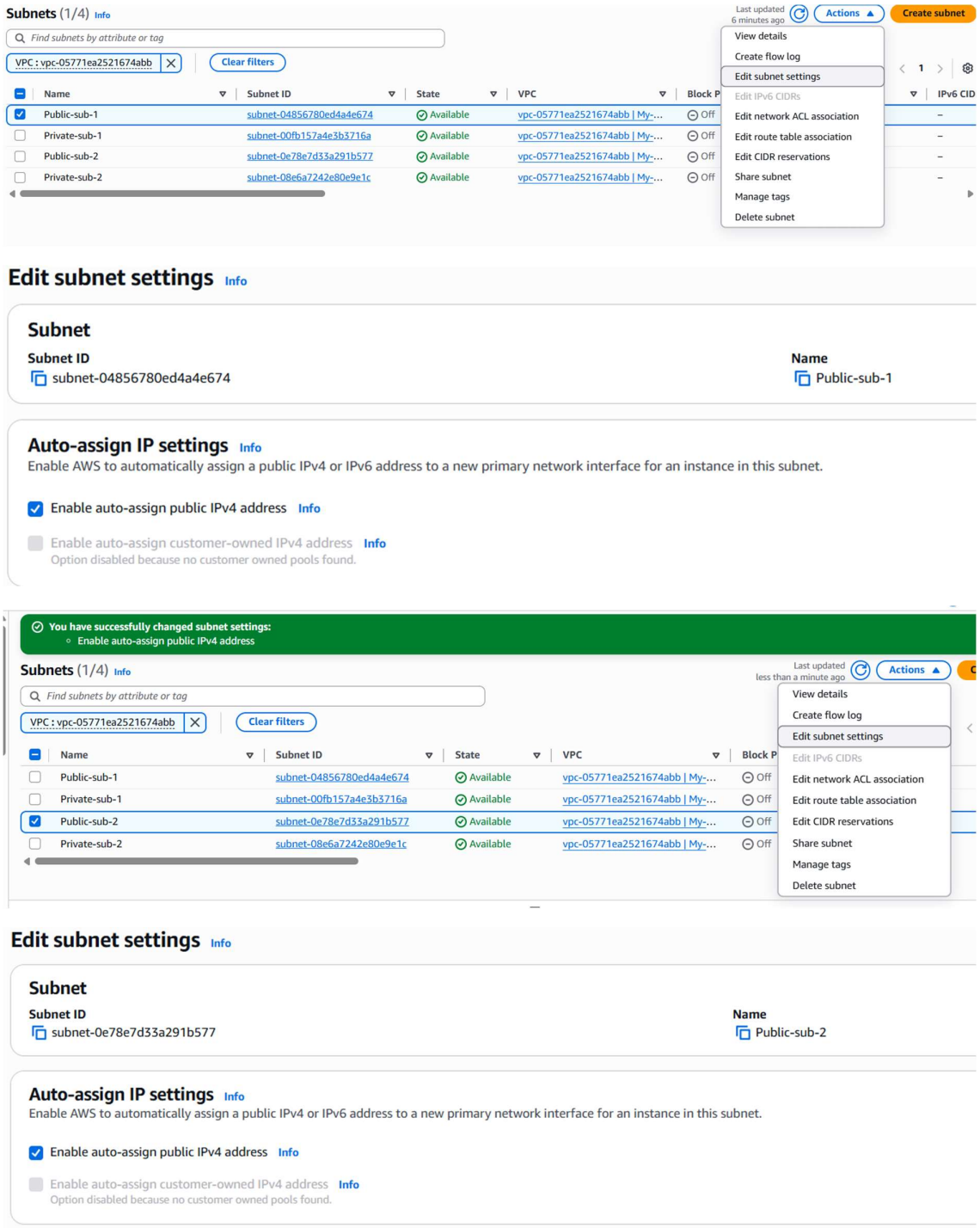


3. Enable Auto Assign Public IP in 2 public subnets.

Step 1:- Navigate to Subnets section. Select the public subnet then go to actions and select subnet settings

Step 2:- In subnet settings enable Auto assign IPv4. Then save the changes.

Step 3:- Do the same for second Public Subnet.



4. Add 2 private subnets in private route table.

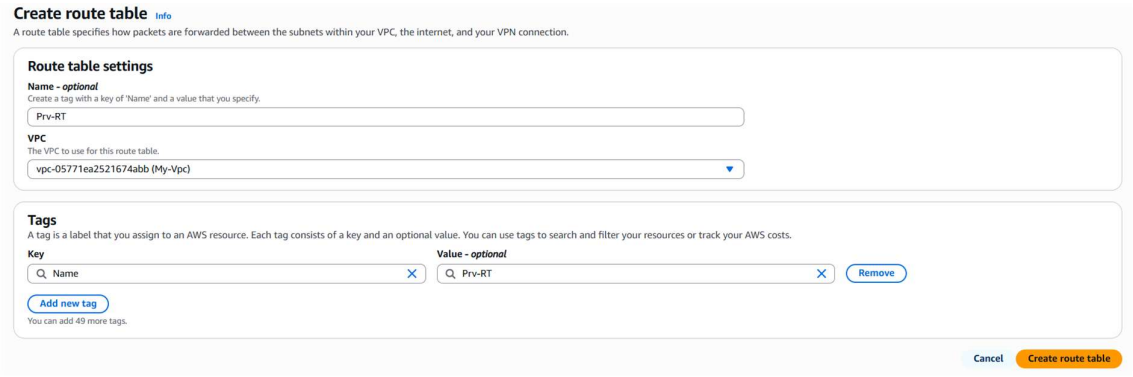
Step 1:- Create Route table with name Prv-RT and use new_VPC in it

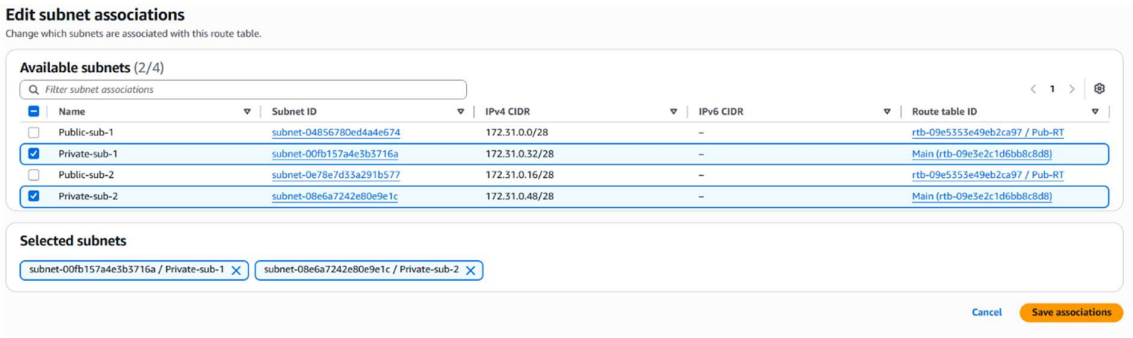
Step 2:- After it is created go to the subnet association section

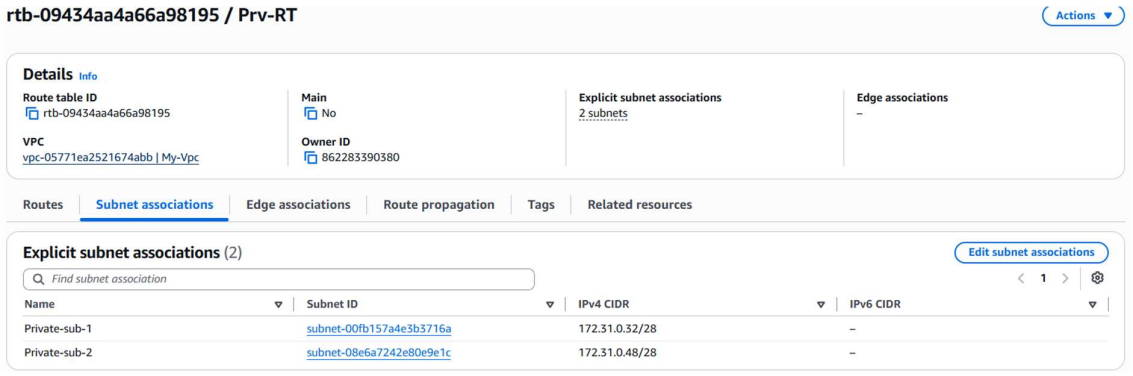
Step 3:- Click on edit associations to add the subnets

Step 4:- Select the private subnets which exists or created before

Step 5:- Save the changes







5. Add 2 public subnets in public route table.

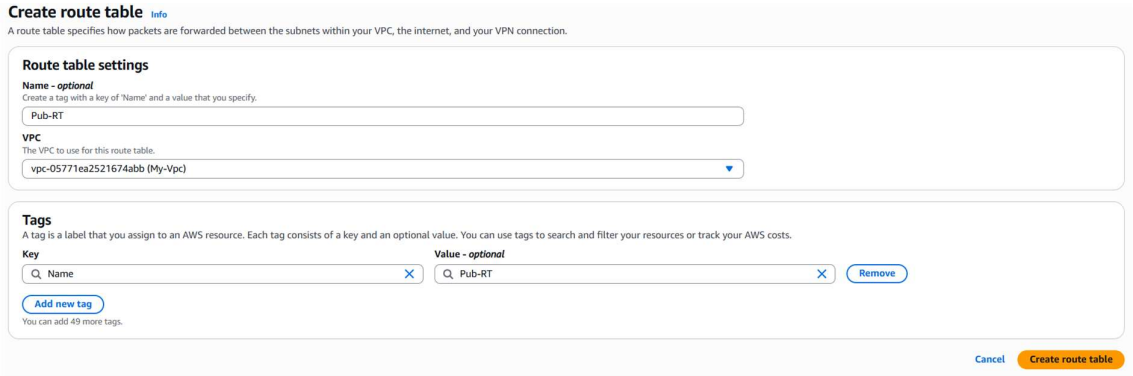
Step 1:- Create Route table with name Pub-RT and use new_VPC in it

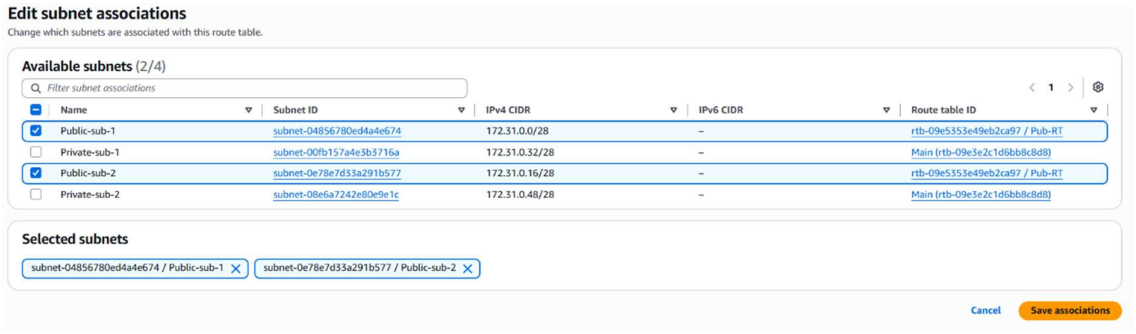
Step 2:- After it is created go to the subnet association section

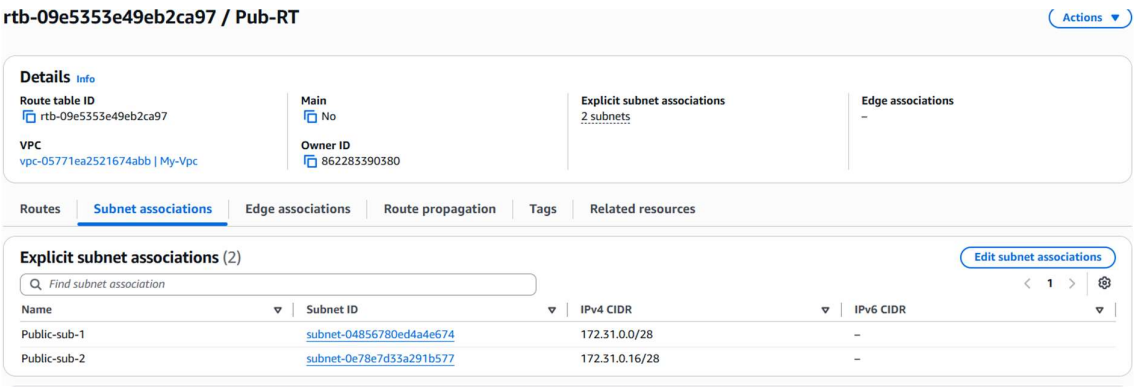
Step 3:- Click on edit associations to add the subnets

Step 4:- Select the public subnets which exists or created before

Step 5:- Save the changes







6. Public route table will have the routes to internet and local.

Step 1:- Create a new Internet Gateway. After creating click actions and go to add VPC

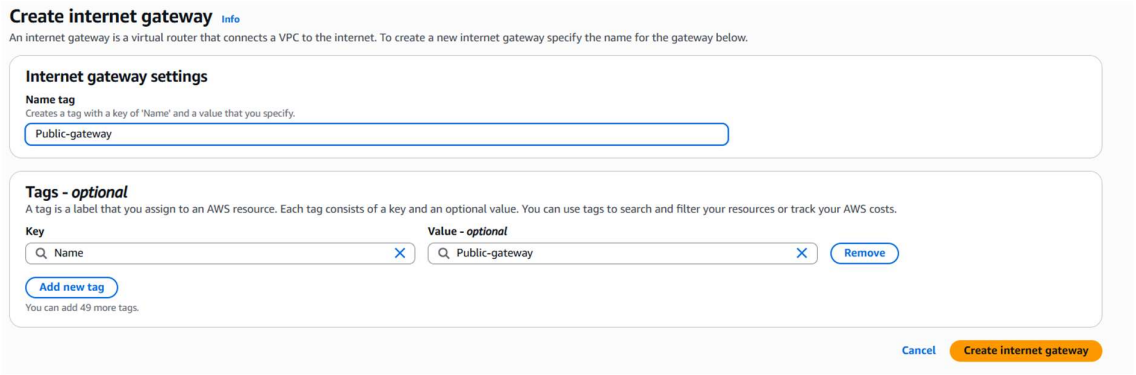
Step 2:- Select the VPC and save Changes.

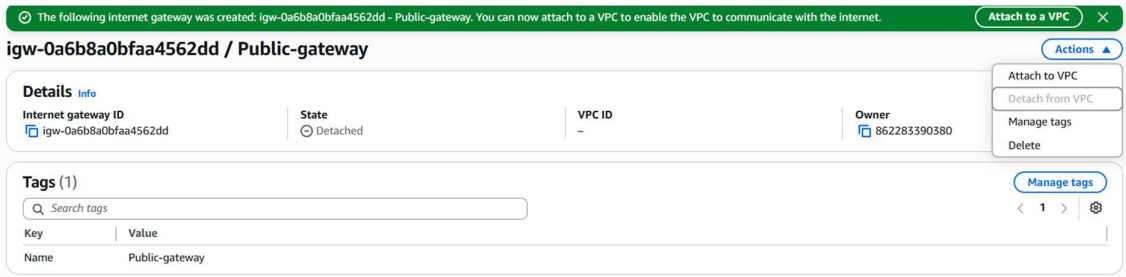
Step 3:- Now go to Route Table section. Select the public RT and go to routes section below.

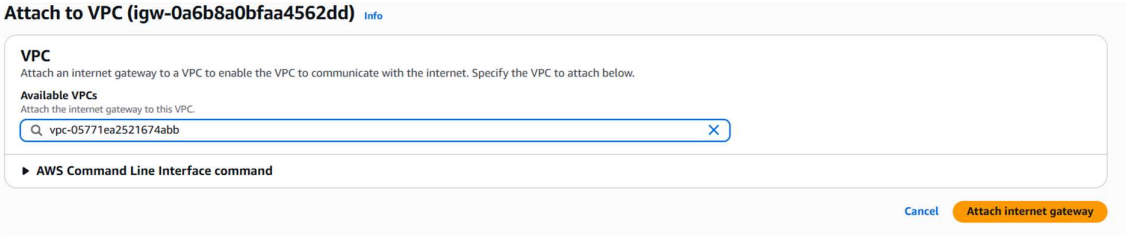
Step 4:- Click on edit routes. Add anywhere IP(0.0.0.0) in the first column and select Internet gateway option in second column.

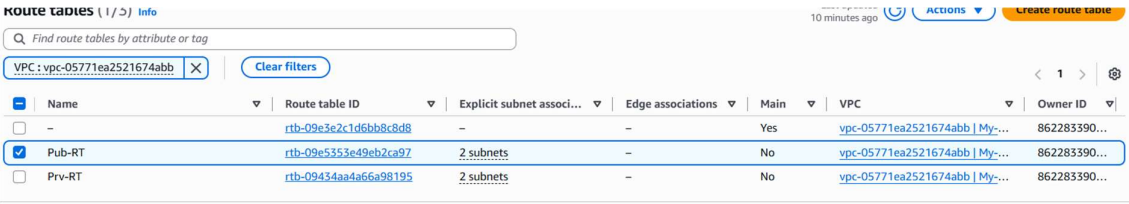
Step 5:- After select IGW we get a dropdown below it. There select the new created IGW id

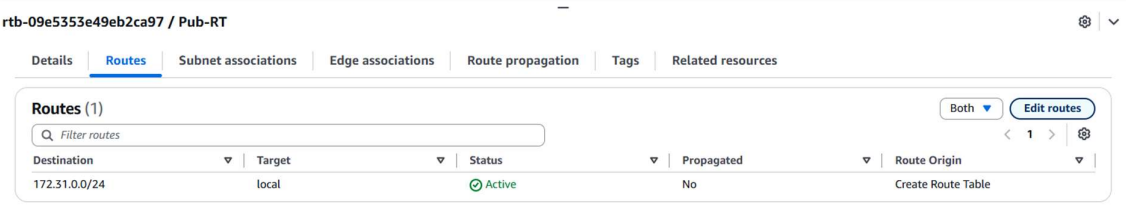
Step 6:- After checking everything save the changes.

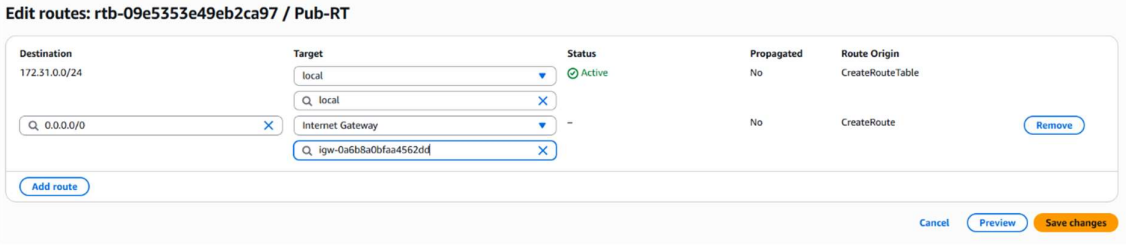


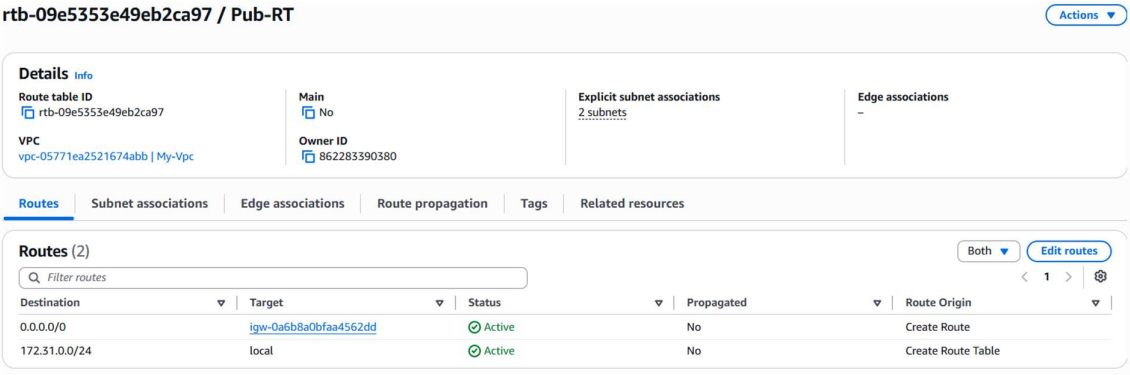












7. Create EC2 in public subnet with t2.micro and install PHP.

Step 1:- Create a instance with t2.micro type. Change the VPC and subnet to Public.

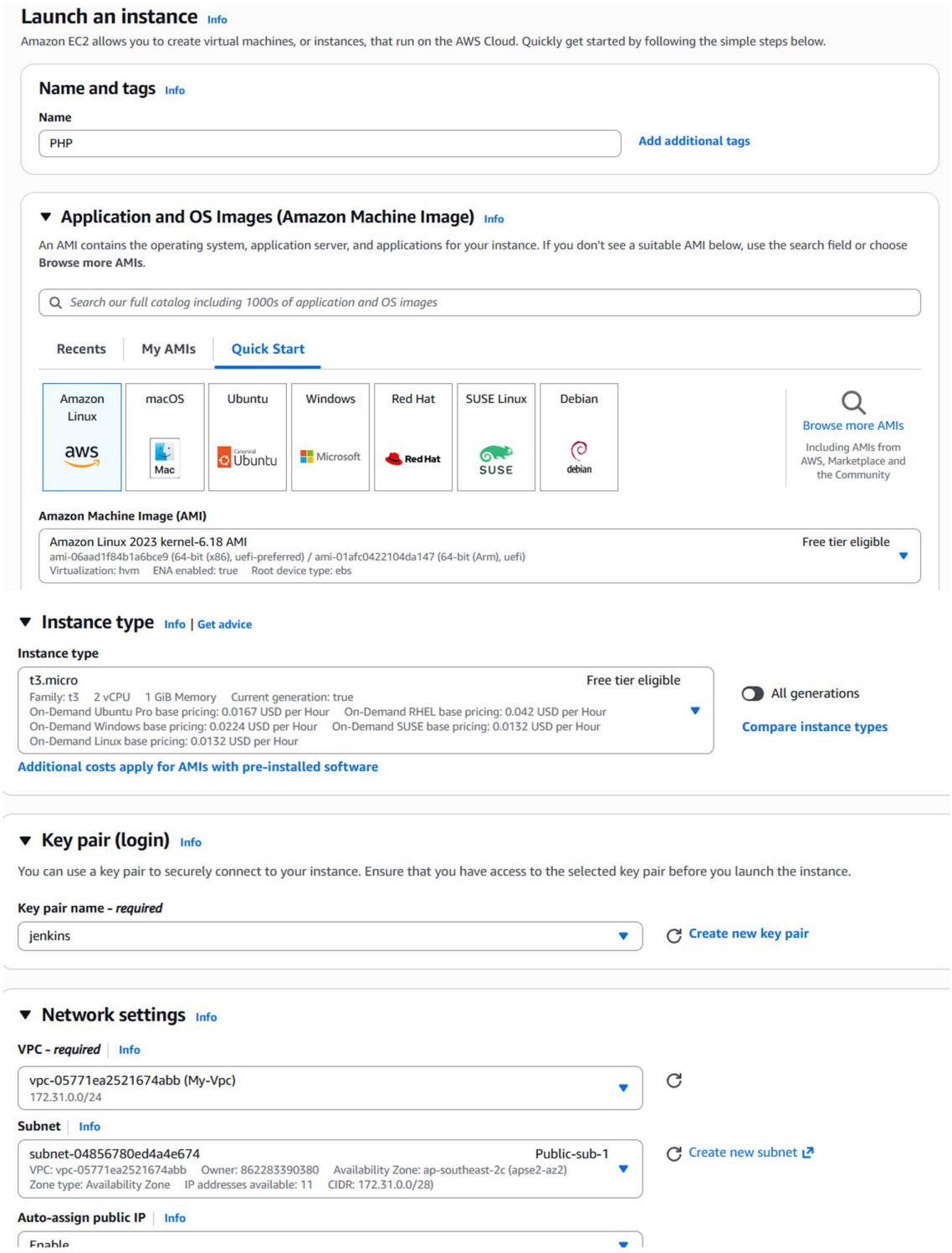
Step 2:- Configure the security add ssh port and all traffic to Anywhere

Step 3:- Then run the instance. Install the PHP package. To confirm the download use php -v

Step 4:- Now start the HTTPD package (Installed as dependency of PHP) with systemctl.

Step 5:- Now create a custom php page and save it at /var/www/html folder

Step 6:- Then open the web-browser use public-ip of instance with php file then we can see our custom page.(public-ip/index.php)



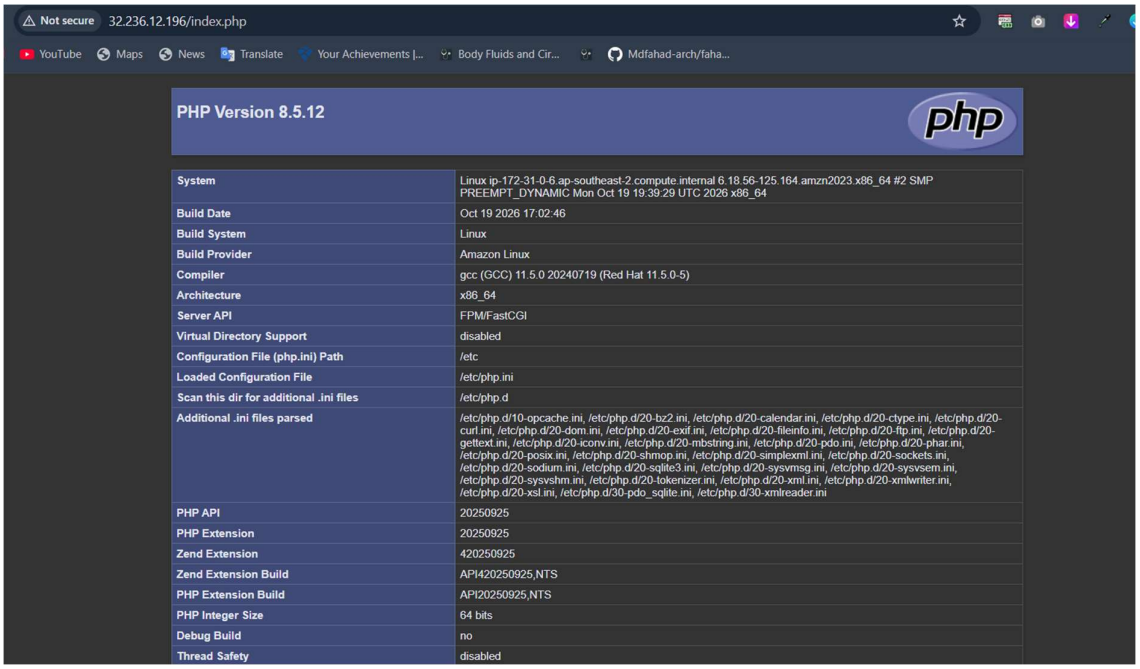
```
[ec2-user@ip-172-31-0-6 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Last metadata expiration check: 0:00:01 ago on Thu Oct 29 17:48:21 2026.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-0-6 ~]$ sudo yum install php -y
Last metadata expiration check: 0:00:25 ago on Thu Oct 29 17:48:21 2026.
Dependencies resolved.
================================================================================
 Package                          Architecture          Version
================================================================================
Installing:
 php8.5                           x86_64                8.5.12-1.amzn2023.0.1
Installing dependencies:
 apr                              x86_64                1.7.5-1.amzn2023.0.4
 apr-util                         x86_64                1.6.5-1.amzn2023.0.1
 apr-util-lmdb                    x86_64                1.6.5-1.amzn2023.0.1
 generic-logos-httpd              noarch                18.0.0-12.amzn2023.0.3
 httpd-core                       x86_64                2.4.69-1.amzn2023.0.2
 httpd-filesystem                 noarch                2.4.69-1.amzn2023.0.2
 httpd-tools                      x86_64                2.4.69-1.amzn2023.0.2
 libbrotli                        x86_64                1.1.0-1.amzn2023.0.2
 libsodium                        x86_64                1.0.19-5.amzn2023
 mailcap                          noarch                2.1.49-3.amzn2023.0.3
 nginx-filesystem                 noarch                1:1.30.6-1.amzn2023.0.1
 php8.5-cli                       x86_64                8.5.12-1.amzn2023.0.1
 php8.5-common                    x86_64                8.5.12-1.amzn2023.0.1
 php8.5-process                   x86_64                8.5.12-1.amzn2023.0.1
 php8.5-xml                       x86_64                8.5.12-1.amzn2023.0.1
Installing weak dependencies:
 apr-util-openssl                 x86_64                1.6.5-1.amzn2023.0.1
 httpd                            x86_64                2.4.69-1.amzn2023.0.2
 mod_http2                        x86_64                1:2.4.69-1.amzn2023.0.2
 mod_lua                          x86_64                2.4.69-1.amzn2023.0.2
 php8.5-fpm                       x86_64                8.5.12-1.amzn2023.0.1
 php8.5-mbstring                  x86_64                8.5.12-1.amzn2023.0.1
```

```
[root@ip-172-31-0-6 ~]# php -v
PHP 8.5.12 (cli) (built: Oct 19 2026 17:02:46) (NTS gcc x86_64)
Copyright (c) The PHP Group
Built by Amazon Linux
Zend Engine v4.5.12, Copyright (c) Zend Technologies
    with Zend OPcache v8.5.12, Copyright (c), by Zend Technologies
[root@ip-172-31-0-6 ~]#
```

```
[ec2-user@ip-172-31-0-6 ~]$ sudo systemctl start httpd
[ec2-user@ip-172-31-0-6 ~]$ sudo systemctl status httpd
● httpd.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
    Drop-In: /usr/lib/systemd/system/httpd.service.d
             └─php-fpm.conf
     Active: active (running) since Thu 2026-10-29 17:51:43 UTC; 11s ago
       Docs: man:httpd.service(8)
   Main PID: 26514 (httpd)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes served/sec:   0 B/sec"
      Tasks: 177 (limit: 1059)
     Memory: 14.2M
        CPU: 79ms
     CGroup: /system.slice/httpd.service
             ├─26514 /usr/sbin/httpd -DFOREGROUND
             ├─26537 /usr/sbin/httpd -DFOREGROUND
             ├─26538 /usr/sbin/httpd -DFOREGROUND
             ├─26539 /usr/sbin/httpd -DFOREGROUND
             └─26663 /usr/sbin/httpd -DFOREGROUND

Oct 29 17:51:43 ip-172-31-0-6.ap-southeast-2.compute.internal systemd[1]: Starting httpd.service - The Apache
Oct 29 17:51:43 ip-172-31-0-6.ap-southeast-2.compute.internal systemd[1]: Started httpd.service - The Apache
Oct 29 17:51:43 ip-172-31-0-6.ap-southeast-2.compute.internal httpd[26514]: Server configured, listening on:
[ec2-user@ip-172-31-0-6 ~]$
```

```
[root@ip-172-31-0-6 ~]# echo '<?php phpinfo(); ?>' > /var/www/html/index.php
[root@ip-172-31-0-6 ~]#
```



8. Configure NAT gateway in public subnet and connect to private instance.

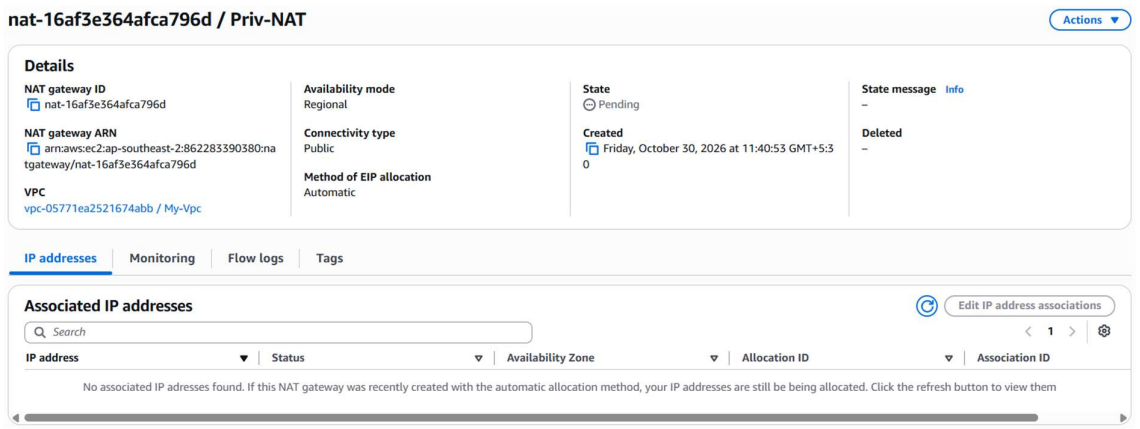
Step 1:- Create a NAT gateway with new_VPC and assign it to private subnets in the route table.

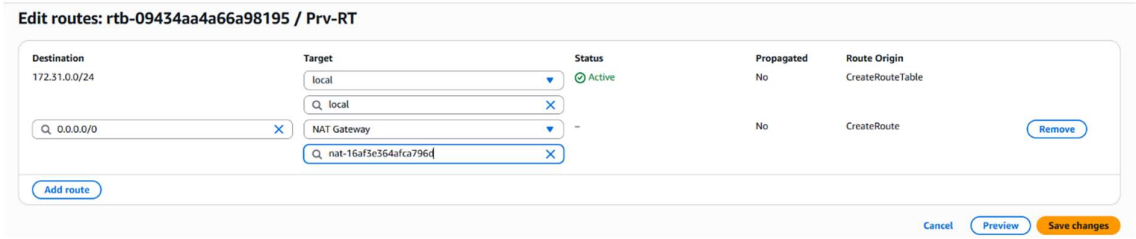
Step 2:- Then create a public instance with public subnet which has IGW and then private instance with private subnet which has NAT

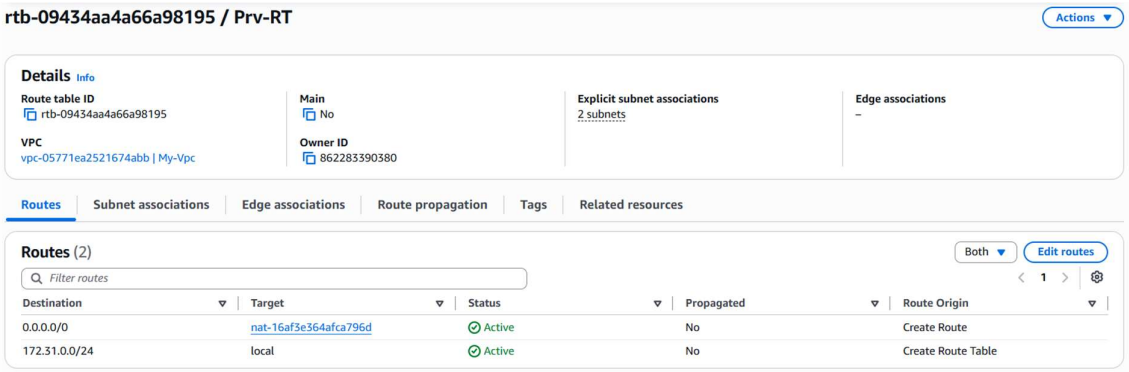
Step 3:- First establish ssh connection with the public instance. Then copy the pem key into instance. Change the permission to 600.

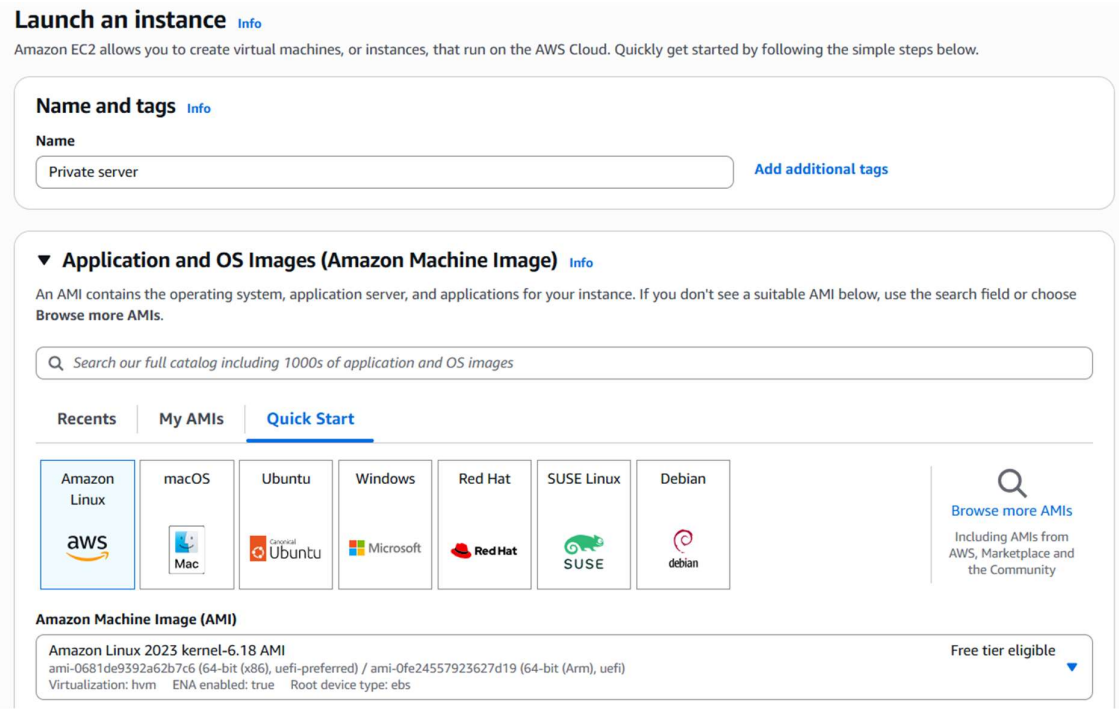
Step 4:- Now using ssh connect with private instace from the public instance. Use private IP of Private instance to establish connection

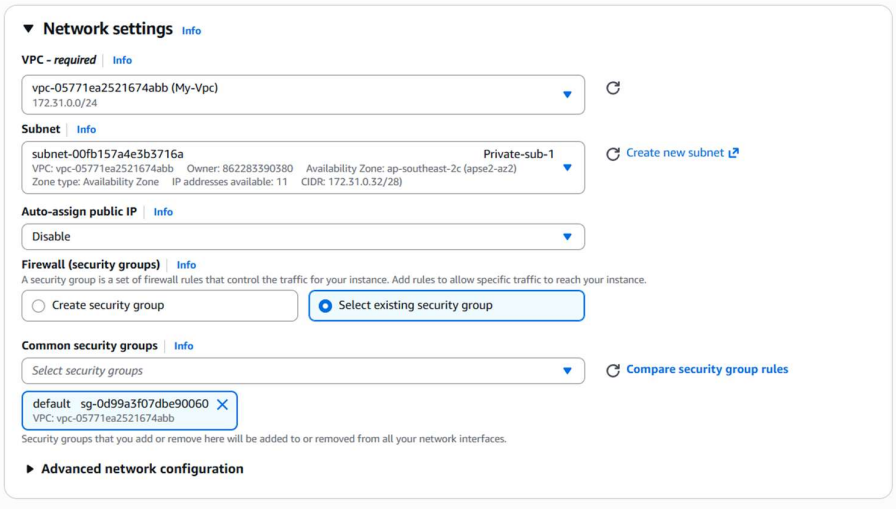
Step 5:- If connected run any command which uses internet to check NAT gateway is working or not.




Edit routes: rtb-09434aa4a66a98195 / Prv-RT
Destination
Target
Status
Propagated
Route Origin
172.31.0.0/24
local
Active
No
CreateRouteTable
local
0.0.0.0/0
NAT Gateway
No
CreateRoute
Remove
nat-16af3e364afca796d
Add route
Cancel
Preview
Save changes


rtb-09434aa4a66a98195 / Prv-RT
Actions
Details Info
Route table ID
rtb-09434aa4a66a98195
Main
No
Explicit subnet associations
2 subnets
Edge associations
–
VPC
vpc-05771ea2521674abb | My-Vpc
Owner ID
862283390380
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
0.0.0.0/0
nat-16af3e364afca796d
Active
No
Create Route
172.31.0.0/24
local
Active
No
Create Route Table


Launch an instance Info
Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.
Name and tags Info
Name
Private server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
My AMIs
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Amazon Linux 2023 kernel-6.18 AMI
Free tier eligible
ami-0681de9392a62b7c6 (64-bit (x86), uefi-preferred) / ami-0fe24557923627d19 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs


Network settings Info
VPC - required Info
vpc-05771ea2521674abb (My-Vpc)
172.31.0.0/24
Subnet Info
subnet-00fb157a4e3b3716a
Private-sub-1
VPC: vpc-05771ea2521674abb Owner: 862283390380 Availability Zone: ap-southeast-2c (apse2-az2)
Zone type: Availability Zone IP addresses available: 11 CIDR: 172.31.0.32/28)
Create new subnet
Auto-assign public IP Info
Disable
Firewall (security groups) Info
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.
Create security group
Select existing security group
Common security groups Info
Select security groups
Compare security group rules
default sg-0d99a3f07dbe90060
VPC: vpc-05771ea2521674abb
Security groups that you add or remove here will be added to or removed from all your network interfaces.
Advanced network configuration

Launch an instance Info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags Info

Name

Public Subnet

Add additional tags

Application and OS Images (Amazon Machine Image) Info

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

Search our full catalog including 1000s of application and OS images

Recents | My AMIs | Quick Start

Amazon Linux | macOS | Ubuntu | Windows | Red Hat | SUSE Linux | Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI — Free tier eligible

ami-0681de9392a62b7c6 (64-bit (x86), uefi-preferred) / ami-0fe24557923627d19 (64-bit (Arm), uefi)

Virtualization: hvm ENA enabled: true Root device type: ebs

Network settings Info

VPC - *required* Info

vpc-05771ea2521674abb (My-Vpc)
172.31.0.0/24

Subnet Info

subnet-04856780ed4a4e674 Public-sub-1

VPC: vpc-05771ea2521674abb Owner: 862283390380 Availability Zone: ap-southeast-2c (apse2-az2)

Zone type: Availability Zone IP addresses available: 10 CIDR: 172.31.0.0/28)

Create new subnet

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

○ Create security group ◉ Select existing security group

Common security groups Info

Select security groups

Compare security group rules

default sg-0d99a3f07dbe90060
VPC: vpc-05771ea2521674abb

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▸ Advanced network configuration

```
Faisal@LAPTOP-4DO8SS4E UCRT64 ~/Downloads
$ ssh -i jenkins.pem ec2-user@16.176.177.70
The authenticity of host '16.176.177.70 (16.176.177.70)' can't be established.
ED25519 key fingerprint is: SHA256:v+gd123qFadTJNHXqXXGgFBrxMEP0II3UtvWi1XKEiO
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '16.176.177.70' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
       #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 30 06:21:10 2026 from 13.239.158.4
[ec2-user@ip-172-31-0-13 ~]$ sudo yum  update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-0-13 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-0-13 ~]$ vi jenkins.pem
[ec2-user@ip-172-31-0-13 ~]$
```

```
ec2-user@ip-172-31-0-13:~
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEA8nSdTKrfqVZqQi2AyMledREi+az/82U+eW9btabMeKDR+2FT
k28NONxNW1VMaeo8+ejDSLEevROi5c2undBvP7kSrd48QcZQP8yFYtqfDZaKLu6R
RaM8hT4Eu7WNiVy655YXscwlX+F+AOVg1OD0OO/lkwFa6BkWqO904hQVgqLV1YGk
u9bagaXBIf1PqfExq9JE/hSUZekRvFbOJaAIn3JbVJOrI/8zgwIy6hknuN75jzJd
mS/gze4vk9eQiRufSJgqxpzsdOfuZ4gBpUhOmCSBxZ21y7eKN3CYW9AYXDMoxHyz
o5zwHmUu1DS5Fh4rnQJdXmCsf49+jcOBlvVK3wIDAQABAoIBAQDYU2gOAd/xjSLd
zQ+9Fr5qoVGdh9gO6OSnIM6o8xw4/PSd4CbbwxiGpCl5NlApqE+jgyTKdQ5CGf8L
HOuPj2vTZWPfMgx9e14arnbMztdfHlRIr596jLzw/XbnrIOObehtvSODBezK8byn
Wn2fUVueGI2zbmAX+lqZmxlkphrTJI4kjIidyaH8QMXr1oUSq8Pzaazd43d/8hjT
HBp819TBoAbLapV1yBcf6Ug17uVXaji2B7rYKmpn5u/xZGpCyAmgBclM2wkLOUsu
NcMaHkd+JlUFWvLVftTP5GYbwAcQa1HB2c/cfh1JBOnwNpvFJZaIvvzvxJ75qeOP
BkmOZtFBAoGBAPpdlrsGZPi89sf9AmLyi1bimGGYQb0gAKelFjSck/yjtnIvAfFO
DTJ3VG6LkGlJJjecypkVJkUNwuDmPGAJvsekUzgQU6/NleYUS7EW9fiiU3ekPXwm
hgpvYIS/ZTvnqCf3+pwxgpAVmKiLk1CgtNtFGgmNK2Ah8LnHhZSMuUxPAoGBAPfp
dD8CMviIOu73nzx/AWBhX1zjuZ4SkHTs7i5XA4Ltd48wlArcxullRMghCSU9VSp+
Jk3CGn8MCMYOizYR4n4aLjqqkaONDiSFj6YYF56AOU9bXgAUylfQ2TqVXAZeTOuE
htjtJJFwjpbGbOniN3Mcs4wmcze5+9cCY8r9OKRxAoGBAOO8SVumz36y79TI9b30
vfvycxnnQEOJU5WHZOcthzzZlEPWLyOMHIO5bZPiQ1DAs6n2tDMiSOUbqRGamJMY
KOLb3s89CU+IqwV9Lcv8Sc7zYcv4oRpHFrREnVysT6zX8gsNezcN4v3MuP7cYxb/
cOBlNirezb8l6gjf+r+sKNkZAoGACZXnPsBxBBcWQNMmPwqSHQ3myAquFKSXHU5D
71rB8gAIjWrNbpYWwtN9k1djqSLz9g6moH8OOza96BwD/fU3ErUkcHRD6bMFhvf7
E+r+jsxaau0Xd11WbYrPu2onAsAGxN688ApLTaxSdxX5G5w5lKPZdWPPNKpmSGv5
O9IKrKECgYEAiyYxJZVpZOSU2kDBjVz6SHeJuIdoTiZ85vZ+zM5fbdlMka6mZZOr
NwuwbkH5zEN3UzQ12KCyTjQ3gGnxS949SbVcn+j/Gfu/dUsOHIwOn+A8d6G2uT8d
XOe7dlSO8gL4P73cHZDUrk4z1eqmkRKJwCw035EGKDB4mEVKvDQsJUQ=
-----END RSA PRIVATE KEY-----
```

```
[ec2-user@ip-172-31-0-13 ~]$ ssh -i jenkins.pem ec2-user@54.252.164.115
ssh: connect to host 54.252.164.115 port 22: Connection timed out
[ec2-user@ip-172-31-0-13 ~]$ ssh -i jenkins.pem ec2-user@172.31.0.40
The authenticity of host '172.31.0.40 (172.31.0.40)' can't be established.
ED25519 key fingerprint is SHA256:YjOptzMOPMmfKrHclbAOupALazDnE5va36Pu1X6tk3o.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.31.0.40' (ED25519) to the list of known hosts.
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
@         WARNING: UNPROTECTED PRIVATE KEY FILE!          @
@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@@
Permissions 0644 for 'jenkins.pem' are too open.
It is required that your private key files are NOT accessible by others.
This private key will be ignored.
Load key "jenkins.pem": bad permissions
ec2-user@172.31.0.40: Permission denied (publickey,gssapi-keyex,gssapi-with-mic).
[ec2-user@ip-172-31-0-13 ~]$ sudo chmod 400 jenkins.pem
[ec2-user@ip-172-31-0-13 ~]$ ssh -i jenkins.pem ec2-user@172.31.0.40
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-0-40 ~]$
```

```
[ec2-user@ip-172-31-0-13 ~]$ ssh -i jenkins.pem ec2-user@172.31.0.40
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
[ec2-user@ip-172-31-0-40 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-0-40 ~]$
```

9. Install Apache Tomcat in private EC2 and deploy a sample app.

Step 1:- Install the tomcat package with wget. Also install java

Step 2:- Then create a service file at /system/system path

Step 3:- the use daemon-reload to initialize service file then start the package.

Step 4:- Check the package is running or not with systemctl.

Step 5:- we cannot open web-browser because it is private instance and doesn't have internet access. So we use curl to get response.

Step 6:- Now go to webapps folder in tomcat folder. Create a folder and a html file in it.

Step 7:- Use curl to get this response.

Step 8:- If we want to see it in web-browser we have to configure it in cmd terminal.

Step 9:- We have to change the permissions of the pem file.

```
[ec2-user@ip-172-31-0-40 opt]$ sudo wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
--2026-10-30 06:43:43--  https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.27/bin/apache-tomcat-11.0.27.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14545574 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.27.tar.gz'

apache-tomcat-11.0.27.tar.gz                      100%[=============================================================================

2026-10-30 06:43:46 (353 MB/s) - 'apache-tomcat-11.0.27.tar.gz' saved [14545574/14545574]

[ec2-user@ip-172-31-0-40 opt]$ ls
apache-tomcat-11.0.27.tar.gz  aws
[ec2-user@ip-172-31-0-40 opt]$ sudo tar -zxvf apache-tomcat-11.0.27.tar.gz
apache-tomcat-11.0.27/conf/
apache-tomcat-11.0.27/conf/catalina.properties
apache-tomcat-11.0.27/conf/context.xml
apache-tomcat-11.0.27/conf/jaspic-providers.xml
apache-tomcat-11.0.27/conf/jaspic-providers.xsd
apache-tomcat-11.0.27/conf/logging.properties
apache-tomcat-11.0.27/conf/server.xml
apache-tomcat-11.0.27/conf/tomcat-users.xml
apache-tomcat-11.0.27/conf/tomcat-users.xsd
apache-tomcat-11.0.27/conf/web.xml
apache-tomcat-11.0.27/bin/
apache-tomcat-11.0.27/lib/
apache-tomcat-11.0.27/logs/
apache-tomcat-11.0.27/temp/
apache-tomcat-11.0.27/webapps/
apache-tomcat-11.0.27/webapps/ROOT/
apache-tomcat-11.0.27/webapps/ROOT/WEB-INF/
apache-tomcat-11.0.27/webapps/docs/
apache-tomcat-11.0.27/webapps/docs/META-INF/
apache-tomcat-11.0.27/webapps/docs/WEB-INF/
apache-tomcat-11.0.27/webapps/docs/WEB-INF/jsp/
apache-tomcat-11.0.27/webapps/docs/annotationapi/
apache-tomcat-11.0.27/webapps/docs/api/
apache-tomcat-11.0.27/webapps/docs/appdev/
```

```
root@ip-172-31-0-40:/opt/tomcat/webapps/sample
[ec2-user@ip-172-31-0-40 system]$ sudo -i
[root@ip-172-31-0-40 ~]# cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Server
After=network.target

[Service]
Type=forking

User=root
Group=root

Environment="JAVA_HOME=/usr/lib/jvm/java-21-amazon-corretto.x86_64"
Environment="CATALINA_HOME=/opt/tomcat"
Environment="CATALINA_BASE=/opt/tomcat"

ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure
RestartSec=10

[Install]
WantedBy=multi-user.target
```

```
[root@ip-172-31-0-40 ~]# systemctl daemon-reload
[root@ip-172-31-0-40 ~]# systemctl start tomcat
[root@ip-172-31-0-40 ~]# systemctl status tomcat
● tomcat.service - Apache Tomcat Web Application Server
     Loaded: loaded (/etc/systemd/system/tomcat.service; enabled; preset: disabled)
     Active: active (running) since Fri 2026-10-30 08:02:10 UTC; 1h 16min ago
   Main PID: 2587 (java)
      Tasks: 29 (limit: 1059)
     Memory: 136.7M
        CPU: 11.483s
     CGroup: /system.slice/tomcat.service
             └─2587 /usr/lib/jvm/java-21-amazon-corretto.x86_64/bin/java -Djava.util.loggi

Oct 30 08:02:10 ip-172-31-0-40.ap-southeast-2.compute.internal systemd[1]: Starting tomcat
Oct 30 08:02:10 ip-172-31-0-40.ap-southeast-2.compute.internal startup.sh[2580]: Tomcat st
Oct 30 08:02:10 ip-172-31-0-40.ap-southeast-2.compute.internal systemd[1]: Started tomcat.
```

```
[root@ip-172-31-0-40 ~]# curl http://localhost:8080



<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8" />
        <title>Apache Tomcat/11.0.27</title>
        <link href="favicon.ico" rel="icon" type="image/x-icon" />
        <link href="tomcat.css" rel="stylesheet" type="text/css" />
    </head>

    <body>
        <div id="wrapper">
            <div id="navigation" class="curved container">
                <span id="nav-home"><a href="https://tomcat.apache.org/">Home</a></span>
                <span id="nav-hosts"><a href="/docs/">Documentation</a></span>
                <span id="nav-config"><a href="/docs/config/">Configuration</a></span>
                <span id="nav-examples"><a href="/examples/">Examples</a></span>
                <span id="nav-wiki"><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Wiki</a></span>
                <span id="nav-lists"><a href="https://tomcat.apache.org/lists.html">Mailing Lists</a></span>
                <span id="nav-help"><a href="https://tomcat.apache.org/findhelp.html">Find Help</a></span>
                <br class="separator" />
            </div>
            <div id="asf-box">
                <h1>Apache Tomcat/11.0.27</h1>
            </div>
            <div id="upper" class="curved container">
                <div id="congrats" class="curved container">
                    <h2>If you're seeing this, you've successfully installed Tomcat. Congratulations!</h2>
                </div>
                <div id="notice">
                    <img id="tomcat-logo" src="tomcat.svg" alt="[tomcat logo]" />
                    <div id="tasks">
                        <h3>Recommended Reading:</h3>
                        <h4><a href="/docs/security-howto.html">Security Considerations How-To</a></h4>
                        <h4><a href="/docs/manager-howto.html">Manager Application How-To</a></h4>
                        <h4><a href="/docs/cluster-howto.html">Clustering/Session Replication How-To</a></h4>
                    </div>
                </div>
                <div id="actions">
                    <div class="button">
                        <a class="container shadow" href="/manager/status"><span>Server Status</span></a>
                    </div>
                    <div class="button">
                        <a class="container shadow" href="/manager/html"><span>Manager App</span></a>
                    </div>
                    <div class="button">
                        <a class="container shadow" href="/host-manager/html"><span>Host Manager</span></a>
                    </div>
```

```
[root@ip-172-31-0-40 ~]# cd /opt/tomcat/webapps/
[root@ip-172-31-0-40 webapps]# ls
ROOT  docs  examples  host-manager  manager
[root@ip-172-31-0-40 webapps]# mkdir sample
[root@ip-172-31-0-40 webapps]# cd sample
[root@ip-172-31-0-40 sample]# touch index.html
[root@ip-172-31-0-40 sample]# chmod 777 index.html
[root@ip-172-31-0-40 sample]# vi index.html
[root@ip-172-31-0-40 sample]# cat index.html
<!DOCTYPE html>
<html>
<head>
    <title>My Sample Application</title>
</head>
<body>
    <h1>Welcome to My Sample Web Application!</h1>
    <p>My application is running successfully on Apache Tomcat.</p>
</body>
</html>
[root@ip-172-31-0-40 sample]# systemctl restart tomcat
```

```
[root@ip-172-31-0-40 sample]# systemctl restart tomcat
[root@ip-172-31-0-40 sample]# curl http://localhost:8080/sample/
<!DOCTYPE html>
<html>
<head>
    <title>My Sample Application</title>
</head>
<body>
    <h1>Welcome to My Sample Web Application!</h1>
    <p>My application is running successfully on Apache Tomcat.</p>
</body>
</html>
[root@ip-172-31-0-40 sample]#
```

```
C:\Windows\System32>icacls "C:\Users\SUBHAN\Downloads\jenkins.pem" /inheritance:r
processed file: C:\Users\SUBHAN\Downloads\jenkins.pem
Successfully processed 1 files; Failed processing 0 files

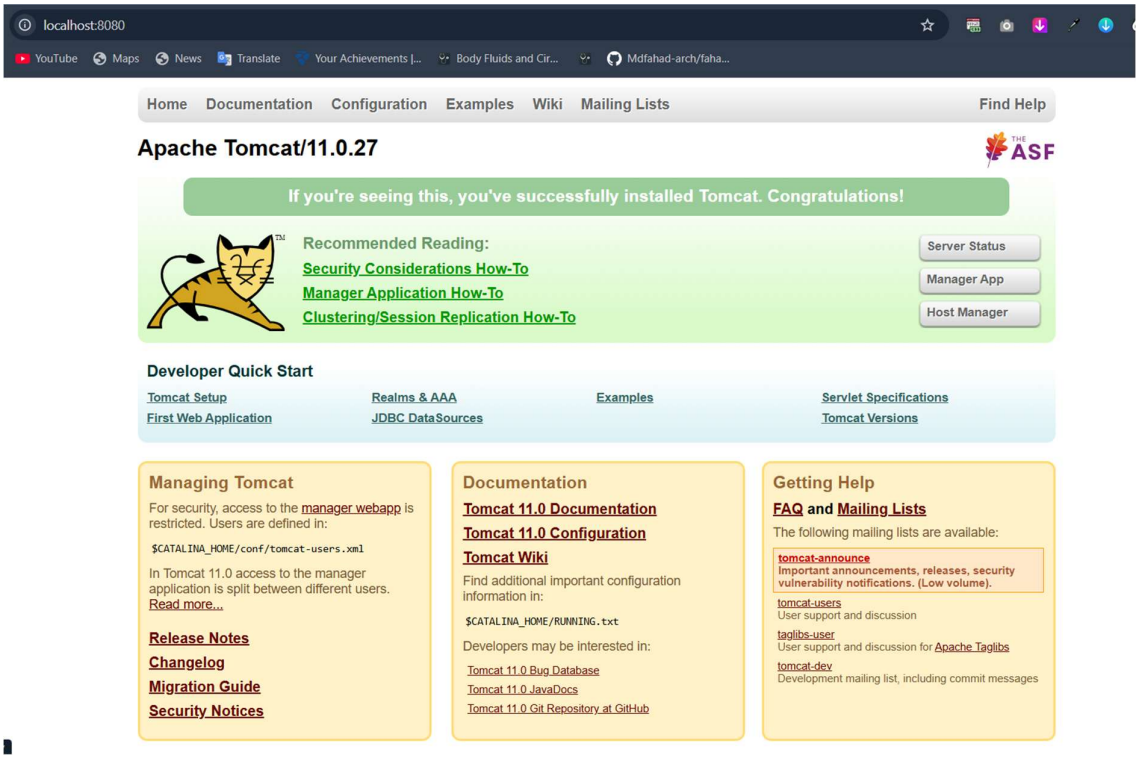
C:\Windows\System32>icacls "C:\Users\SUBHAN\Downloads\jenkins.pem" /grant:r "%USERNAME%:R"
processed file: C:\Users\SUBHAN\Downloads\jenkins.pem
Successfully processed 1 files; Failed processing 0 files

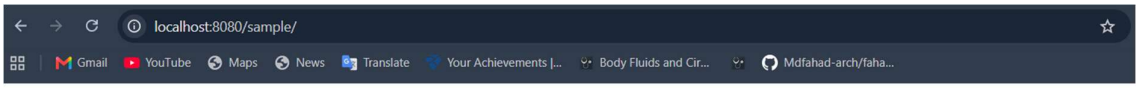
C:\Windows\System32>icacls "C:\Users\SUBHAN\Downloads\jenkins.pem" /remove "Users" "Authenticated Users" "Everyone"
processed file: C:\Users\SUBHAN\Downloads\jenkins.pem
Successfully processed 1 files; Failed processing 0 files

C:\Windows\System32>icacls "C:\Users\SUBHAN\Downloads\jenkins.pem"
C:\Users\SUBHAN\Downloads\jenkins.pem LAPTOP-4DO8SS4E\Faisal:(R)

Successfully processed 1 files; Failed processing 0 files

C:\Windows\System32>ssh -i "C:\Users\SUBHAN\Downloads\jenkins.pem" -L 8080:172.31.0.40:8080 ec2-user@16.176.177.70
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Oct 30 09:13:58 2026 from 103.203.172.230
[ec2-user@ip-172-31-0-13 ~]$ S
```





Welcome to My Sample Web Application!

My application is running successfully on Apache Tomcat.

10. Configure VPC flow logs and store the logs in S3 and CloudWatch.

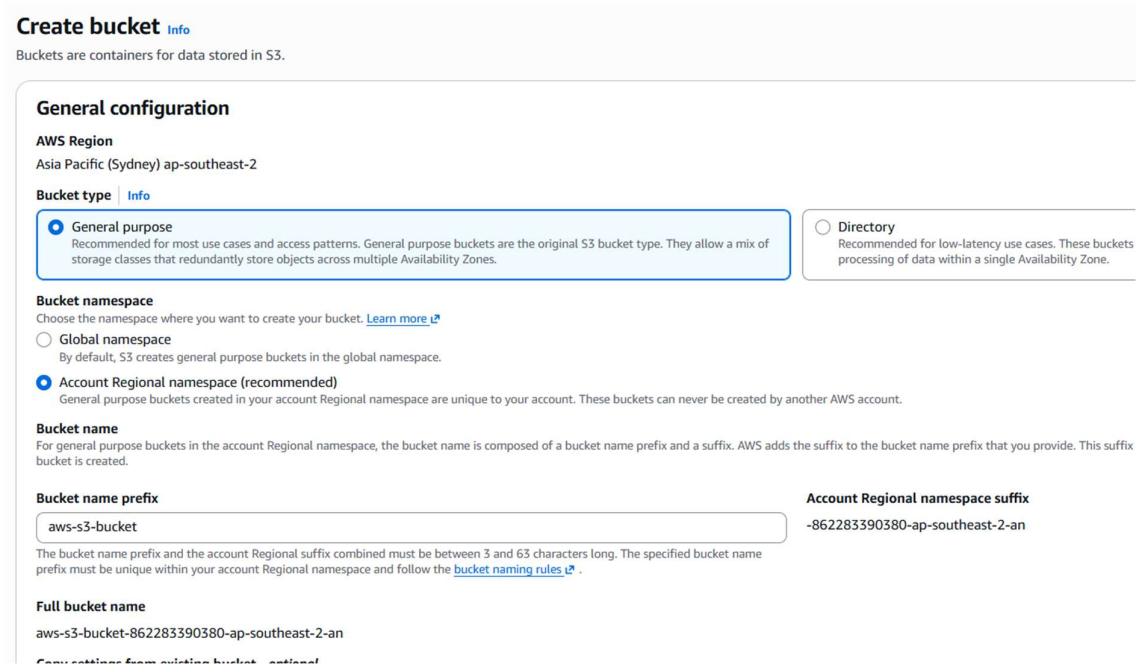
S3 Flow log

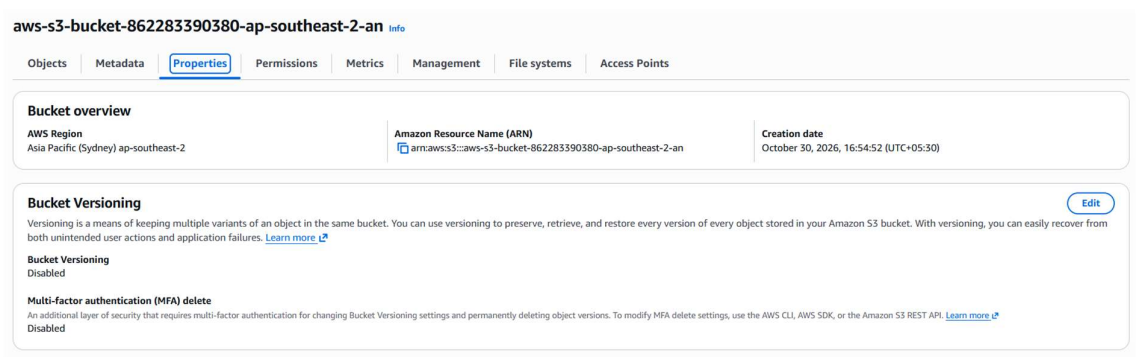
Step 1:- Go to VPC scroll down go to flow logs and create a flow logs.

Step 2:- First create a S3 bucket and get the ARN. Now select create s3 flowlogs in VPC flowlog settings.

Step 3:- paste the S3 ARN in the bucket ARN. Then click save

Step 4:- Now when we go to S3 service we can see aws bucket name we created in VPC flowlog settings.





Selected resources Info

Flow logs will only be created for resources in an available state.

Name	Resource ID	State
My-Vpc	vpc-05771ea2521674abb	Available

Flow log settings

Name - *optional*

aws-s3-logs

Filter

The type of traffic to capture (accepted traffic only, rejected traffic only, or all traffic).

- ○ Accept
- ○ Reject
- ◉ All

Maximum aggregation interval | Info

The maximum interval of time during which a flow of packets is captured and aggregated into a flow log record.

- ○ 10 minutes
- ◉ 1 minute

Destination

The destination to which to publish the flow log data.

- ○ Send to CloudWatch Logs
- ◉ Send to an Amazon S3 bucket
- ○ Send to Amazon Data Firehose in the same account
- ○ Send to Amazon Data Firehose in a different account

S3 bucket ARN

The ARN of the Amazon S3 bucket to which the flow log is published. You can specify a specific folder in the bucket using the arn:aws:s3:::[bucket-name]/[folder_name]/ format.

arn:aws:s3:::aws-s3-bucket-862283390380-ap-southeast-2-an

Create S3 bucket

Must be in the format: arn:aws:s3:::[bucket-name]

fl-0aaf57a7ff2f33d62 / aws-s3-logs

Actions

Details

Flow Log ID fl-0aaf57a7ff2f33d62	**Destination Type** s3	**Traffic Type** All	**File Format** Plain text
Name aws-s3-logs	**Destination Name** aws-s3-bucket-862283390380-ap-southeast-2-an	**Max Aggregation Interval** 1 minute	**Hive Compatible Partitions** Not enabled
State Active	**IAM Role** –	**Log Format** Default	**Partition Logs** Daily
Creation Time Friday, October 30, 2026 at 16:57:03 GMT+5:30	**Cross Account IAM Role** –	**Tag field specifications** -	

Tags | Integrations

Tags

Manage tags

Search tags

Key	Value
Name	aws-s3-logs

aws-s3-bucket-862283390380-ap-southeast-2-an Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems | Access Points

Objects (1)

Copy S3 URI | Copy URL | Downloa

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to ac

Find objects by prefix

Name	Type	Last modified
AWSLogs/	Folder	-

Amazon S3 > Buckets > aws-s3-bucket-862283390380-ap-southeast-2-an > AWSLogs/ > 862283390380/ > vpcflowlogs/ > ap-southeast-2/ > 2026/ > 10/ > 30/

30/

Copy S3 URI

Objects | Properties

Objects (4)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class
862283390380_vpcflowlogs_ap-southeast-2_fl-0aaf57a7ff2f33d62_20261030T1125Z_6d093f91.log.gz	gz	October 30, 2026, 17:01:56 (UTC+05:30)	458.0 B	Standard
862283390380_vpcflowlogs_ap-southeast-2_fl-0aaf57a7ff2f33d62_20261030T1130Z_86d5624b.log.gz	gz	October 30, 2026, 17:01:56 (UTC+05:30)	403.0 B	Standard
862283390380_vpcflowlogs_ap-southeast-2_fl-0aaf57a7ff2f33d62_20261030T1130Z_d45c54d1.log.gz	gz	October 30, 2026, 17:06:56 (UTC+05:30)	754.0 B	Standard
862283390380_vpcflowlogs_ap-southeast-2_fl-0aaf57a7ff2f33d62_20261030T1135Z_0b5ef527.log.gz	gz	October 30, 2026, 17:06:56 (UTC+05:30)	360.0 B	Standard

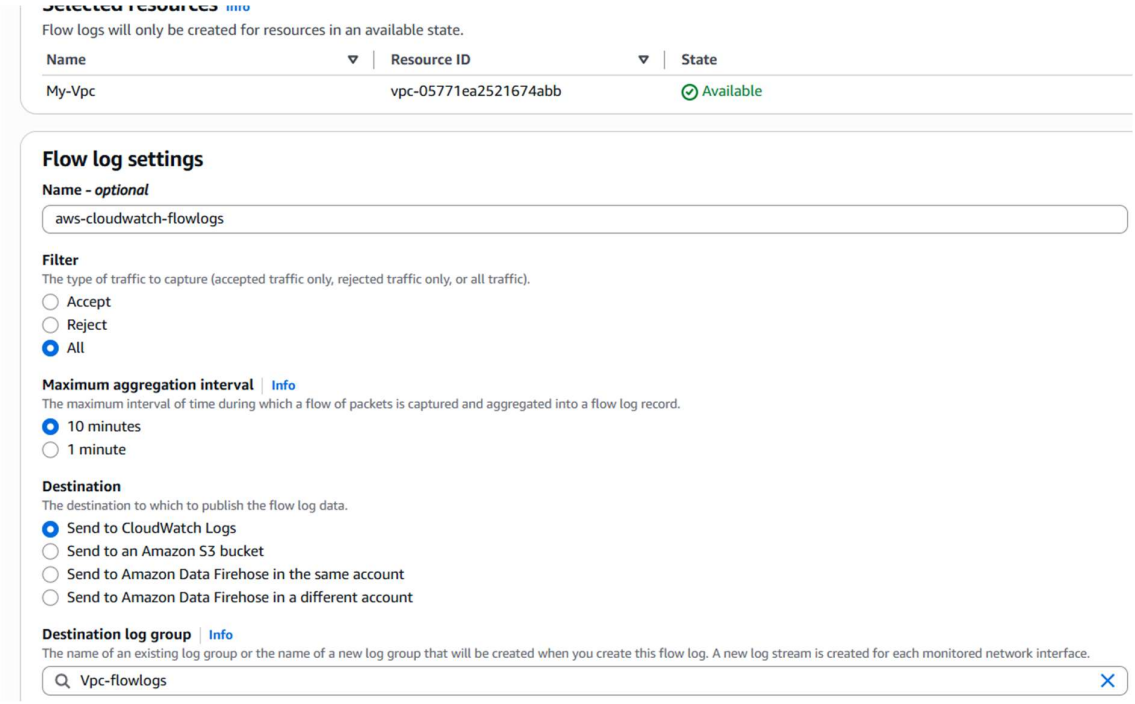
Cloudwatch Flow log

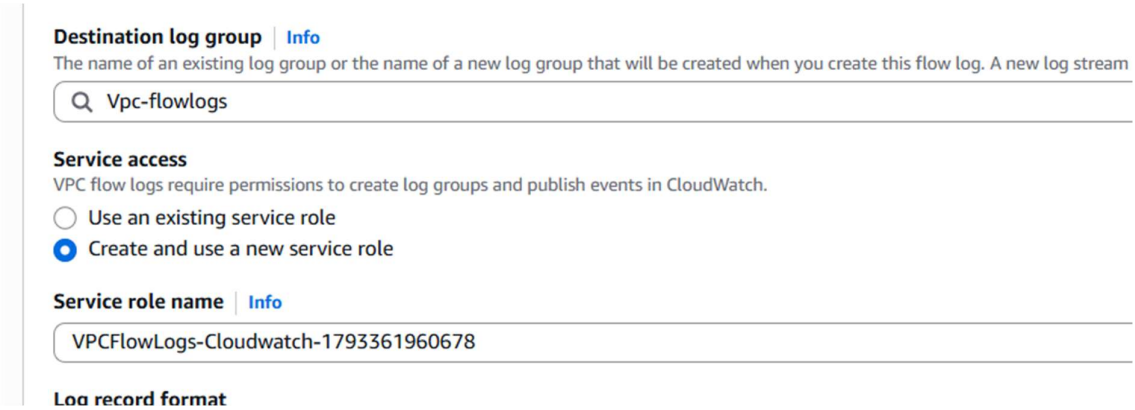
Step 1:- Go to VPC scroll down go to flow logs and create a flow logs.

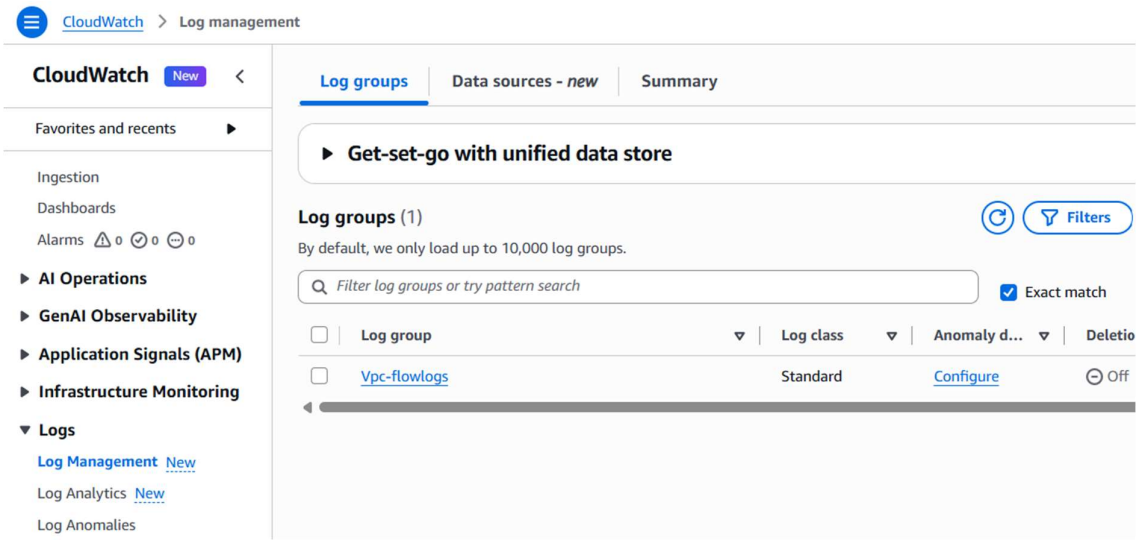
Step 2:- Now select cloudwatch and add a name for log group. Then select create new user.

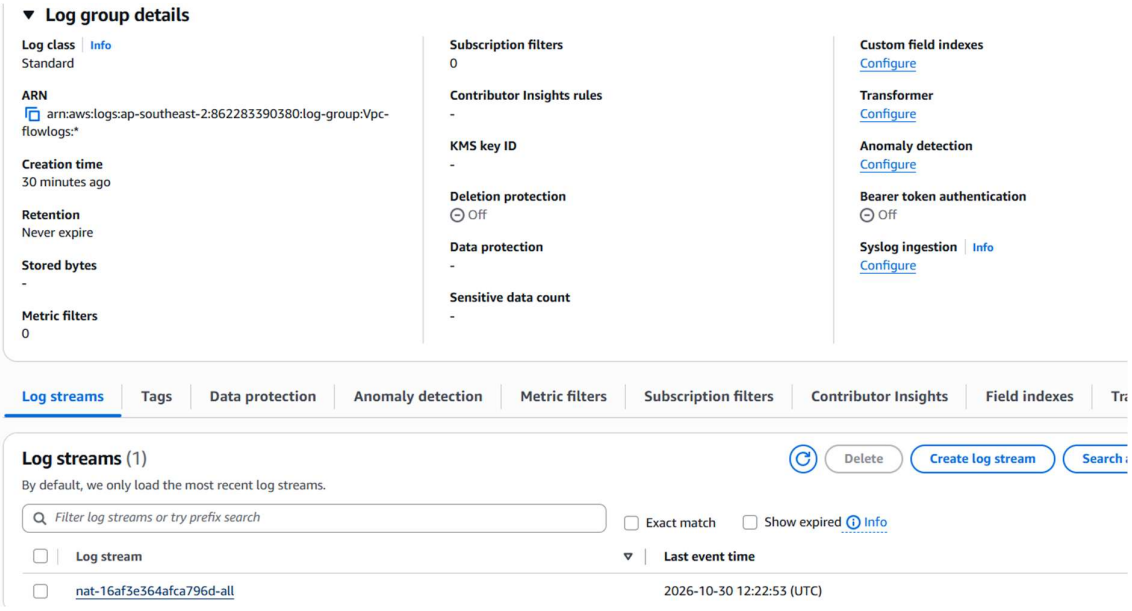
Step 3:- Now save the flow log.

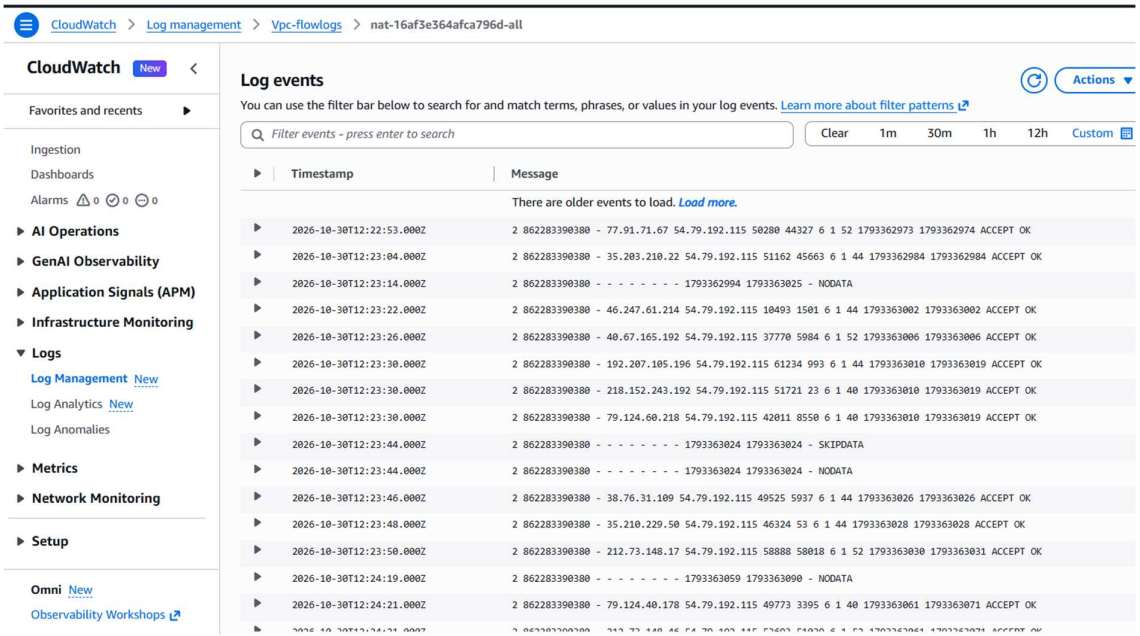
Step 4:- Now when we go to cloudwatch service and go to log management. we can see aws flowlogs we created in VPC flowlog settings.






CloudWatch > Log management
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms 0 0 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Log groups
Data sources - new
Summary
Get-set-go with unified data store
Log groups (1)
By default, we only load up to 10,000 log groups.
Filters
Filter log groups or try pattern search
Exact match
Log group
Log class
Anomaly d...
Deletio
Vpc-flowlogs
Standard
Configure
Off


Log group details
Log class Info
Standard
ARN
arn:aws:logs:ap-southeast-2:862283390380:log-group:Vpc-flowlogs:*
Creation time
30 minutes ago
Retention
Never expire
Stored bytes
-
Metric filters
0
Subscription filters
0
Contributor Insights rules
-
KMS key ID
-
Deletion protection
Off
Data protection
-
Sensitive data count
-
Custom field indexes
Configure
Transformer
Configure
Anomaly detection
Configure
Bearer token authentication
Off
Syslog ingestion Info
Configure
Log streams
Tags
Data protection
Anomaly detection
Metric filters
Subscription filters
Contributor Insights
Field indexes
Log streams (1)
By default, we only load the most recent log streams.
Delete
Create log stream
Filter log streams or try prefix search
Exact match
Show expired Info
Log stream
Last event time
nat-16af3e364afca796d-all
2026-10-30 12:22:53 (UTC)


CloudWatch > Log management > Vpc-flowlogs > nat-16af3e364afca796d-all
CloudWatch New
Favorites and recents
Ingestion
Dashboards
Alarms 0 0 0
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Log Management New
Log Analytics New
Log Anomalies
Metrics
Network Monitoring
Setup
Omni New
Observability Workshops
Log events
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns
Actions
Filter events - press enter to search
Clear
1m
30m
1h
12h
Custom
Timestamp
Message
There are older events to load. Load more.
2026-10-30T12:22:53.000Z 2 862283390380 - 77.91.71.67 54.79.192.115 50280 44327 6 1 52 1793362973 1793362974 ACCEPT OK
2026-10-30T12:23:04.000Z 2 862283390380 - 35.203.210.22 54.79.192.115 51162 45663 6 1 44 1793362984 1793362984 ACCEPT OK
2026-10-30T12:23:14.000Z 2 862283390380 - - - - - - - - 1793362994 1793363025 - NODATA
2026-10-30T12:23:22.000Z 2 862283390380 - 46.247.61.214 54.79.192.115 10493 1501 6 1 44 1793363002 1793363002 ACCEPT OK
2026-10-30T12:23:26.000Z 2 862283390380 - 40.67.165.192 54.79.192.115 37770 5984 6 1 52 1793363006 1793363006 ACCEPT OK
2026-10-30T12:23:30.000Z 2 862283390380 - 192.207.105.196 54.79.192.115 61234 993 6 1 44 1793363010 1793363019 ACCEPT OK
2026-10-30T12:23:30.000Z 2 862283390380 - 218.152.243.192 54.79.192.115 51721 23 6 1 40 1793363010 1793363019 ACCEPT OK
2026-10-30T12:23:30.000Z 2 862283390380 - 79.124.60.218 54.79.192.115 42011 8550 6 1 40 1793363010 1793363019 ACCEPT OK
2026-10-30T12:23:44.000Z 2 862283390380 - - - - - - - - 1793363024 1793363024 - SKIPDATA
2026-10-30T12:23:44.000Z 2 862283390380 - - - - - - - - 1793363024 1793363024 - NODATA
2026-10-30T12:23:46.000Z 2 862283390380 - 38.76.31.109 54.79.192.115 49525 5937 6 1 44 1793363026 1793363026 ACCEPT OK
2026-10-30T12:23:48.000Z 2 862283390380 - 35.210.229.50 54.79.192.115 46324 53 6 1 44 1793363028 1793363028 ACCEPT OK
2026-10-30T12:23:50.000Z 2 862283390380 - 212.73.148.17 54.79.192.115 58888 58018 6 1 52 1793363030 1793363031 ACCEPT OK
2026-10-30T12:24:19.000Z 2 862283390380 - - - - - - - - 1793363059 1793363090 - NODATA
2026-10-30T12:24:21.000Z 2 862283390380 - 79.124.40.178 54.79.192.115 49773 3395 6 1 40 1793363061 1793363071 ACCEPT OK